Sourav Ray

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Professional Summary

Results-driven Data & AI Engineer with 3+ years of enterprise experience building production data platforms, high-throughput ETL/ELT pipelines, and Generative AI systems. Proven expertise in Snowflake, AWS, PySpark, dbt, Apache Airflow, and Multi-Agent Orchestration (MCP). Targeting the AI/ML Engineer role at Codemonk.

Technical Skills

Cloud & Data Platforms: AWS (S3, Lambda, Kinesis, Firehose, DMS, CloudWatch), Snowflake (Snowpark, Snowpipe, Cortex, Zero-Copy Cloning, Time Travel, Cost Optimization), GCP (Vertex AI, BigQuery).
Data Engineering: DBT, Apache Airflow, Apache Spark / PySpark, Coalesce, Apache Kafka, Apache Iceberg, SQL, ETL/ELT, Data Contracts, Data Mesh, Bronze/Silver/Gold Architecture.
AI / ML & GenAI: Large Language Models (LLMs), Agentic AI, Multi-Agent Orchestration, Retrieval-Augmented Generation (RAG), ChromaDB, Vector Search, Model Context Protocol (MCP), LangChain, LangGraph, TensorFlow, NLP, Prompt Engineering.
Programming & Frameworks: Python, SQL, Shell Scripting, FastAPI, Docker, Kubernetes, Git, GitHub Actions, CI/CD.
BI & Tools: Sigma, SAP BusinessObjects validation, Terraform, Linux.

Professional Experience

Factspan Analytics, Bengaluru, India
Senior Analyst / Data & AI Engineer — Sep 2023 – Present

Customer 360 Data Mesh & Governance (Petabyte-Scale) | AWS, Snowflake, DBT, Python

- Architected a petabyte-scale Customer 360 data platform integrating 1,000+ sources across five ingestion patterns: RabbitMQ messaging queues, RDBMS databases, flat files, managed APIs, and real-time/event-based systems.
- Designed the end-to-end ingestion architecture: source systems → AWS ingestion services → Raw Kinesis → Lambda-based data-contract validation → validated Kinesis → Kinesis Firehose → S3 → Snowflake Snowpipe → Bronze/Silver/Gold transformation layers.
- Built a data-contract-driven validation framework that evaluated incoming records against schema and contract requirements before allowing data into the Bronze layer, detecting schema drift, missing mandatory columns, unexpected attributes, and invalid/null values for non-optional fields.
- Implemented fault-tolerant validation and recovery workflows where Lambda processing failures were routed to a messaging queue for controlled retry and reprocessing, preventing transient failures from causing data loss and enabling failed records to re-enter the validation stream.
- Led large-scale data modeling and transformation across approximately 100–200 DBT/SQL models spanning Bronze, Silver, and Gold layers for cross-domain Customer 360 data.
- Designed Bronze-layer transformations to flatten complex nested source payloads into downstream-consumable structures while preserving validated source data.
- Developed Silver-layer transformations for deduplication, enrichment, standardization, and business-level data preparation.

AI-Powered Network Incident Orchestrator | Python, MCP, Multi-Agent, Observability

- Owned the technical architecture for a multi-agent orchestration system coordinating specialized AI agents across ServiceNow, Infoblox, NetBox, SevOne, Datadog, Splunk, and LogicMonitor.
- Wrote and maintained the Python codebase for per-platform MCP servers; conducted design and code reviews with peers on style, testability, and efficiency.
- Designed a per-platform agent architecture, each backed by its own MCP server, providing a consistent abstraction layer over heterogeneous observability tools and device-management APIs, including secure SSH-based access to Cisco and non-Cisco network devices.
- Architected dynamic agent-selection and parallel-invocation logic correlating multi-source data (incidents, logs, metrics, topology, device health, configuration) into a single LLM-synthesized RCA report.

GenAI Schema Validator & Automated Column Mapping | Python, Vector Search, LLMs

- Architected an enterprise-scale schema and report-matching platform combining Exact Match, Semantic Match, and LLM-based inference layers — a RAG-style retrieval-and-reasoning pipeline applied to structured data for automated column mapping.
- Built responsible-AI governance controls to identify and exclude masked PII attributes from LLM inputs.
- Applied prompt engineering to auto-generate metadata for undocumented fields, reducing incorrect field mappings through structured validation.

Education

National Institute of Technology (NIT) Durgapur | M.Tech, Operations Research | CGPA: 8.36/10 — 2021 – 2023
Government College of Engineering and Textile Technology (GCETTB) | B.Tech — 2015 – 2019

Certifications

- **Snowflake:** SnowPro Advanced Architect, SnowPro Advanced Data Scientist
- **AWS Cloud:** AWS Certified Machine Learning – Specialty, AWS Certified Solutions Architect – Associate